

LAMPIRAN A - DOKUMENTASI KODE SUMBER (BAGIAN 3)

Komponen React (Lanjutan) & Flutter Mobile App

File: `react-dashboard/src/components/ManagementPage.jsx`

Deskripsi: Halaman manajemen user (khusus Admin) untuk mengubah role, melihat status online, dan menghapus akses user.

```
import React, { useState, useEffect, useMemo } from "react";
import { ref, onValue, set, remove } from "firebase/database";
import { database } from "../firebase";
import { useAuth } from "../contexts/AuthContext";
import { IoPeopleOutline, IoTrashOutline, IoShieldCheckmarkOutline, IoEllipseSharp } from "react-icons/io5";
import toast from "react-hot-toast";

const ROLE_OPTIONS = [
  { value: "guess", label: "Guess", desc: "Dashboard only" },
  { value: "view_data", label: "View Data", desc: "Dashboard + Data" },
  { value: "eksekusi_data", label: "Eksekusi Data", desc: "Dashboard + Control + Data" },
  { value: "report_data", label: "Report Data", desc: "Dashboard + Data + Export" },
  { value: "admin", label: "Admin", desc: "Full Access" },
];

export default function ManagementPage() {
  const { currentUser } = useAuth();
  const [users, setUsers] = useState({});
  const [userAkses, setUserAkses] = useState({});
  const [presence, setPresence] = useState({});
  const [searchQuery, setSearchQuery] = useState("");
```

```

useEffect(() => {
    const unsub = onValue(ref(database, "/users"), (snap) => setUsers(snap.exists() ? snap.val() : {}));
    return () => unsub();
}, []);

useEffect(() => {
    const unsub = onValue(ref(database, "/user_akses"), (snap) => setUserAkses(snap.exists() ? snap.val() : {}));
    return () => unsub();
}, []);

useEffect(() => {
    const unsub = onValue(ref(database, "/presence"), (snap) => setPresence(snap.exists() ? snap.val() : {}));
    return () => unsub();
}, []);

const userList = useMemo(() => {
    const list = [];
    Object.entries(users).forEach(([uid, userData]) => {
        const akses = userAkses[uid] || {};
        list.push({
            uid, email: userData.email || "N/A",
            displayName: userData.displayName || "-",
            role: akses.role || "guess",
            lastLogin: userData.lastLogin || null,
            isSelf: uid === currentUser?.uid,
        });
    });
    if (searchQuery) {
        const q = searchQuery.toLowerCase();
        return list.filter((u) => u.email.toLowerCase().includes(q) || u.displayName.toLowerCase().includes(q) || u.role.toLowerCase().inclu
    }
    return list;
}, [users, userAkses, currentUser, searchQuery]);

const handleRoleChange = async (uid, newRole) => {
    try {
        await set(ref(database, `user_akses/${uid}`), { role: newRole, updatedAt: Date.now() });
        toast.success("Role berhasil diperbarui!");
    }

```

```

    } catch (err) { toast.error("Gagal memperbarui role."); }
  };

  const handleDeleteAccess = async (uid, email) => {
    if (!window.confirm(`Hapus akses untuk ${email}?`)) return;
    try {
      await remove(ref(database, `user_akses/${uid}`));
      toast.success("Akses berhasil dihapus.");
    } catch (err) { toast.error("Gagal menghapus akses."); }
  };

  // ... render table with user list, role select, delete button
}

```

File: **react-dashboard/src/App.jsx** (Komponen Utama)

Deskripsi: Komponen utama yang mengelola state global, tab navigation, listener Firebase, dan rendering semua halaman (Dashboard, Control, Data, Management).

Fitur Utama dalam App.jsx:

- Listener real-time ke 4 node Firebase: **/sensor_data** , **/blowers** , **/control_mode** , **/history**
- Tab navigation dengan filter RBAC berdasarkan role user
- Status koneksi (online/offline) berdasarkan timestamp terakhir (threshold 120 detik)
- Normalisasi data 10 sensor ke array tetap
- Master blower control (mengatur semua blower sekaligus)
- Tabel data historis dengan search, pagination, dan sorting
- Export CSV dengan dukungan native (Capacitor) dan web (Blob download)

*// Kode Lengkap App.jsx terdapat di file asli:
 // react-dashboard/src/App.jsx (596 baris)
 // Lihat Lampiran A1 untuk referensi kode ESP32*

File: react-dashboard/capacitor.config.ts

Deskripsi: Konfigurasi Capacitor untuk build APK Android dari React web app.

```
import type { CapacitorConfig } from '@capacitor/cli';

const config: CapacitorConfig = {
  appId: 'com.solardryer.iot',
  appName: 'Solar Dryer IoT',
  webDir: 'build',
  android: {
    backgroundColor: '#fffcf2',
    allowMixedContent: true,
  },
  server: {
    androidScheme: 'https',
    url: 'https://solardryeriot.netlify.app',
    cleartext: true,
  },
};

export default config;
```

File: react-dashboard/database.rules.json

Deskripsi: Aturan keamanan Firebase Realtime Database.

```
{
  "rules": {
    "sensor_data": { ".read": true, ".write": true },
    "blowers": { ".read": true, ".write": true },
    "control_mode": { ".read": true, ".write": true },
```

```
"history": { ".read": true, ".write": true },
"users": {
  ".read": "auth != null && root.child('user_akses').child(auth.uid).child('role').val() === 'admin'",
  "$uid": {
    ".read": "auth != null && auth.uid === $uid",
    ".write": "auth != null && auth.uid === $uid"
  }
},
"user_akses": {
  ".read": "auth != null && root.child('user_akses').child(auth.uid).child('role').val() === 'admin'",
  "$uid": {
    ".read": "auth != null && auth.uid === $uid",
    ".write": "auth != null && (root.child('user_akses').child(auth.uid).child('role').val() === 'admin' || !data.exists())"
  }
}
}
```

3. FLUTTER MOBILE APP

File: flutter_dashboard/lib/main.dart

Deskripsi: Entry point aplikasi Flutter dengan tema dark mode Material3.

```
import 'package:flutter/material.dart';
import 'package:firebase_core/firebase_core.dart';
import 'package:google_fonts/google_fonts.dart';
import 'screens/dashboard_screen.dart';

void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp();
  runApp(const SmartACApp());
}
```

```
}
```

```
class SmartACApp extends StatelessWidget {  
  const SmartACApp({super.key});  
  @override  
  Widget build(BuildContext context) {  
    return MaterialApp(  
      title: 'Smart AC Dashboard',  
      debugShowCheckedModeBanner: false,  
      theme: ThemeData(  
        useMaterial3: true,  
        brightness: Brightness.dark,  
        scaffoldBackgroundColor: const Color(0xFF0F172A),  
        textTheme: GoogleFonts.interTextTheme(ThemeData.dark().textTheme),  
        colorScheme: ColorScheme.fromSeed(seedColor: const Color(0xFF3B82F6), brightness: Brightness.dark),  
      ),  
      home: const DashboardScreen(),  
    );  
  }  
}
```

File: flutter_dashboard/lib/services/firebase_service.dart

Deskripsi: Layanan akses Firebase Realtime Database untuk Flutter.

```
class FirebaseService {  
  final DatabaseReference _db = FirebaseDatabase.instance.ref();  
  
  Stream<DatabaseEvent> get sensorDataStream =>  
    _db.child('sensor_data').onValue;  
  
  Stream<DatabaseEvent> get blowersStream =>  
    _db.child('blowers').onValue;
```

```

Stream<DatabaseEvent> get historyStream =>
    _db.child('history').orderByChild('timestamp').limitToLast(1000).onValue;

Stream<DatabaseEvent> get controlModeStream =>
    _db.child('control_mode').onValue;

Future<void> setBlowerSpeed(String blowerId, int speed) async {
    await _db.child('blowers/$blowerId/speed').set(speed);
}

Future<void> setControlMode(String mode) async {
    await _db.child('control_mode').set(mode);
}
}
\n```

---

### File: `flutter_dashboard/lib/widgets/sensor_card.dart`
**Deskripsi**: Widget kartu sensor dengan indikator warna status dinamis.

```dart
import 'package:flutter/material.dart';

class SensorCard extends StatelessWidget {
 final String type;
 final double? value;
 final String unit;

 const SensorCard({super.key, required this.type, required this.value, required this.unit});

 bool get isTemp => type == 'temperature';

 Color get statusColor {
 if (value == null) return Colors.grey;
 if (isTemp) {
 if (value! > 35) return Colors.red;
 if (value! > 28) return Colors.orange;
 return Colors.green;
 }
 }
}

```

```

 }
 if (value! > 80) return Colors.red;
 if (value! > 60) return Colors.orange;
 return Colors.green;
}

```

```

String get statusText {
 if (value == null) return '--';
 if (isTemp) {
 if (value! > 35) return 'Panas!';
 if (value! > 28) return 'Hangat';
 return 'Normal';
 }
 if (value! > 80) return 'Lembab!';
 if (value! > 60) return 'Sedang';
 return 'Normal';
}

```

```

@override
Widget build(BuildContext context) {
 return Container(
 padding: const EdgeInsets.all(20),
 decoration: BoxDecoration(
 gradient: const LinearGradient(colors: [Color(0xFF1E293B), Color(0xFF162032)]),
 borderRadius: BorderRadius.circular(20),
 border: Border.all(color: const Color(0xFF334155)),
),
 child: Row(children: [
 Icon(isTemp ? Icons.thermostat_rounded : Icons.water_drop_rounded, size: 48, color: statusColor),
 const SizedBox(width: 16),
 Expanded(child: Column(crossAxisAlignment: CrossAxisAlignment.start, children: [
 Text(isTemp ? 'SUHU' : 'KELEMBABAN', style: TextStyle(fontSize: 12, color: Colors.grey[500])),
 Text(value != null ? value!.toStringAsFixed(1) : '--',
 style: TextStyle(fontSize: 36, fontWeight: FontWeight.w800, color: statusColor)),
 Container(
 padding: const EdgeInsets.symmetric(horizontal: 10, vertical: 3),
 decoration: BoxDecoration(color: statusColor.withValues(alpha: 0.15), borderRadius: BorderRadius.circular(50)),
 child: Text(statusText, style: TextStyle(fontSize: 11, fontWeight: FontWeight.w600, color: statusColor)),
),
]),
]),
),

```

```
])),
]),
);
}
}
```

---

**File:** flutter\_dashboard/lib/widgets/blower\_control.dart

---

**Deskripsi:** Widget slider untuk mengontrol kecepatan blower via Firebase.

```
import 'package:fluttertoast/fluttertoast.dart';
import '../services/firebase_service.dart';

class BlowerControl extends StatefulWidget {
 final String id;
 final String name;
 final int speed;
 final bool isDisabled;

 const BlowerControl({
 super.key,
 required this.id,
 required this.name,
 required this.speed,
 this.isDisabled = false,
 });

 @override
 State<BlowerControl> createState() => _BlowerControlState();
}

class _BlowerControlState extends State<BlowerControl> {
 final _firebase = FirebaseService();
 double _localSpeed = 0;
 bool _isDragging = false;
```

```

@override
void initState() {
 super.initState();
 _localSpeed = widget.speed.toDouble();
}

@override
void didUpdateWidget(BlowerControl oldWidget) {
 super.didUpdateWidget(oldWidget);
 if (!_isDragging && oldWidget.speed != widget.speed) {
 _localSpeed = widget.speed.toDouble();
 }
}

Future<void> _updateSpeed(double val) async {
 final int finalSpeed = val.round();
 try {
 await _firebase.setBlowerSpeed(widget.id, finalSpeed);
 FlutternToast.showToast(
 msg: '${widget.name} speed set to $finalSpeed%',
 backgroundColor: Colors.amber[800],
 textColor: Colors.white,
);
 } catch (e) {
 FlutternToast.showToast(
 msg: 'Gagal mengatur kecepatan!',
 backgroundColor: Colors.red,
);
 }
}

@override
Widget build(BuildContext context) {
 int rpm = ((_localSpeed / 100) * 2480).round();
 final bool disabled = widget.isDisabled;

 return Container(
 padding: const EdgeInsets.all(20),

```

```
decoration: BoxDecoration(
 gradient: const LinearGradient(
 begin: Alignment.topLeft,
 end: Alignment.bottomRight,
 colors: [
 Color(0xFF1E293B),
 Color(0xFF162032),
],
),
 borderRadius: BorderRadius.circular(20),
 border: Border.all(color: const Color(0xFF334155)),
),
child: Column(
 crossAxisAlignment: CrossAxisAlignment.start,
 children: [
 Text(
 'AIRFLOW MANAGEMENT',
 style: TextStyle(
 fontSize: 10,
 fontWeight: FontWeight.w700,
 color: Colors.grey[500],
 letterSpacing: 1.2,
),
),
 const SizedBox(height: 8),
 Row(
 mainAxisAlignment: MainAxisAlignment.spaceBetween,
 crossAxisAlignment: CrossAxisAlignment.end,
 children: [
 Column(
 crossAxisAlignment: CrossAxisAlignment.start,
 children: [
 Text(
 widget.name,
 style: const TextStyle(
 fontSize: 22,
 fontWeight: FontWeight.bold,
 color: Color(0xFFE2E8F0),
 height: 1.1,
```

```

),
),
 const Text(
 'Control',
 style: TextStyle(
 fontSize: 22,
 fontWeight: FontWeight.bold,
 color: Color(0xFFE2E8F0),
 height: 1.1,
),
),
),
],
),
Column(
 crossAxisAlignment: CrossAxisAlignment.end,
 children: [
 if (disabled)
 Container(
 padding: const EdgeInsets.symmetric(horizontal: 8, vertical: 3),
 margin: const EdgeInsets.only(bottom: 4),
 decoration: BoxDecoration(
 color: Colors.amber.withValues(alpha: 0.15),
 borderRadius: BorderRadius.circular(6),
),
),
 child: const Text(
 'AUTO',
 style: TextStyle(
 fontSize: 10,
 fontWeight: FontWeight.w800,
 color: Color(0xFFD97706),
 letterSpacing: 1,
),
),
),
),
Text(
 rpm.toString(),
 style: TextStyle(
 fontSize: 28,
 fontWeight: FontWeight.w500,

```

```

 color: Colors.amber[600],
 height: 1,
),
),
 Text(
 'RPM',
 style: TextStyle(
 fontSize: 12,
 fontWeight: FontWeight.w600,
 color: Colors.grey[500],
),
),
],
),
],
),
const SizedBox(height: 20),
SliderTheme(
 data: SliderThemeData(
 activeTrackColor: Colors.amber[600],
 inactiveTrackColor: const Color(0xFF334155),
 thumbColor: Colors.amber[400],
 overlayColor: Colors.amber.withOpacity(0.2),
 trackHeight: 18,
 thumbShape: const RoundSliderThumbShape(
 enabledThumbRadius: 16,
 elevation: 4,
 pressedElevation: 6,
),
),
),
child: Slider(
 value: _localSpeed,
 min: 0,
 max: 100,
 onChangeStart: disabled ? null : (_) => setState(() => _isDragging = true),
 onChanged: disabled ? null : (val) {
 setState(() {
 _localSpeed = val;
 });
 },

```

```

 },
 onChangeEnd: disabled ? null : (val) {
 setState(() => _isDragging = false);
 _updateSpeed(val);
 },
),
),
Padding(
 padding: const EdgeInsets.symmetric(horizontal: 16.0),
 child: Row(
 mainAxisAlignment: MainAxisAlignment.spaceBetween,
 children: const [
 Text('OFF',
 style: TextStyle(
 fontSize: 10,
 fontWeight: FontWeight.bold,
 color: Colors.grey)),
 Text('50%',
 style: TextStyle(
 fontSize: 10,
 fontWeight: FontWeight.bold,
 color: Colors.grey)),
 Text('MAX',
 style: TextStyle(
 fontSize: 10,
 fontWeight: FontWeight.bold,
 color: Colors.grey)),
],
),
),
],
),
);
}
}
\n```

```

---

### File: `flutter\_dashboard/lib/widgets/chart\_widget.dart`

**\*\*Deskripsi\*\*:** Widget grafik garis menggunakan fl\_chart untuk visualisasi data historis sensor.

```
`` dart

import 'package:flutter/material.dart';
import 'package:fl_chart/fl_chart.dart';

class SensorChartWidget extends StatelessWidget {
 final List<Map<String, dynamic>> data;
 const SensorChartWidget({super.key, required this.data});

 @override
 Widget build(BuildContext context) {
 return Container(
 padding: const EdgeInsets.all(20),
 decoration: BoxDecoration(
 gradient: const LinearGradient(colors: [Color(0xFF1E293B), Color(0xFF162032)]),
 borderRadius: BorderRadius.circular(20),
),
 child: Column(crossAxisAlignment: CrossAxisAlignment.start, children: [
 const Text('Grafik Sensor (1 Jam Terakhir)', style: TextStyle(fontSize: 16, fontWeight: FontWeight.w600)),
 const SizedBox(height: 20),
 data.isEmpty
 ? const SizedBox(height: 200, child: Center(child: Text('Belum ada data historis...')))
 : SizedBox(height: 220, child: LineChart(_buildChart())),
]),
);
 }

 LineChartData _buildChart() {
 final tempSpots = <FlSpot>[];
 final humSpots = <FlSpot>[];
 for (int i = 0; i < data.length; i++) {
 tempSpots.add(FlSpot(i.toDouble(), (data[i]['temperature'] as num?)?.toDouble() ?? 0));
 humSpots.add(FlSpot(i.toDouble(), (data[i]['humidity'] as num?)?.toDouble() ?? 0));
 }
 return LineChartData(
 lineBarsData: [
 LineChartBarData(spots: tempSpots, isCurved: true, color: Colors.orange, barWidth: 2.5),
```

```
 LineChartBarData(spots: humSpots, isCurved: true, color: Colors.blue, barWidth: 2.5),
],
);
 }
}
```

---

**File:** `flutter_dashboard/lib/screens/dashboard_screen.dart`

---

**Deskripsi:** Layar utama dashboard Flutter yang menampilkan sensor, blower control, dan grafik.

```
// Kode Lengkap: 269 baris
// Mendengarkan stream: sensor_data, blowers, history
// Heartbeat timer: cek koneksi setiap 5 detik (threshold 120 detik)
// Layout: SliverAppBar + SliverList dengan SensorCard, BlowerControl, SensorChartWidget
```

---

## 4. KONFIGURASI PROJECT

---

**File:** `react-dashboard/package.json`

---

```
{
 "name": "smart-ac-dashboard",
 "version": "1.0.0",
 "dependencies": {
 "@capacitor/android": "^8.2.0",
 "@capacitor/core": "^8.2.0",
 "@capacitor/filesystem": "^8.1.2",
 "@capacitor/share": "^8.0.1",
 "firebase": "^10.14.1",
 "react": "^18.3.1",
 "react-dom": "^18.3.1",
```

```
 "react-hot-toast": "^2.4.1",
 "react-icons": "^5.3.0",
 "react-scripts": "5.0.1",
 "recharts": "^2.13.3"
 }
}
```

## File: flutter\_dashboard/pubspec.yaml

---

```
name: smart_ac_dashboard
description: Solar Dryer IoT - Flutter Mobile App
version: 1.0.0+1
environment:
 sdk: ">=3.0.0 <4.0.0"
dependencies:
 flutter:
 sdk: flutter
 firebase_core: ^3.8.1
 firebase_database: ^11.2.1
 fl_chart: ^0.69.2
 google_fonts: ^6.2.1
 fluttertoast: ^8.2.8
```